University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

Instructor: Hadi Salmasian

Midterm 1 – October 11, 2011

Version A

Surname _____ First Name _____

Student # _____ DGD Number _____

Instructions:

(1) You have 80 minutes to complete this exam.
(2) The number of points available for each question is indicated in between parentheses.
(3) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(4) Write your student number at the top of each page in the space provided.
(5) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(6) You are strongly recommended to write in **pen**, not pencil.
(7) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	2	4	2	5	20
Score							

1. **(3 points)** Find all solutions to the vector equation

$$x_1 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ -6 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -\frac{3}{5} \end{bmatrix}$$

and write the solution in vector parametric form.

2. **(4 points)** Determine if the following linear system is consistent or inconsistent. If it is consistent, then find its general solution.

$$\begin{cases} -3x + 6y - z + w - 7v = 7 \\ x - 2y + 2z + 3w - v = 1 \\ 2x - 4y + 5z + 8w - 4v = 4 \end{cases}$$

3. **(2 points)** Find the value(s) of h for which the following system has a unique solution:

$$\begin{cases} x + y = 5 \\ 2x + hy = 3 \end{cases}$$

You should justify your answer.

4. **(4 points)** Suppose that

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

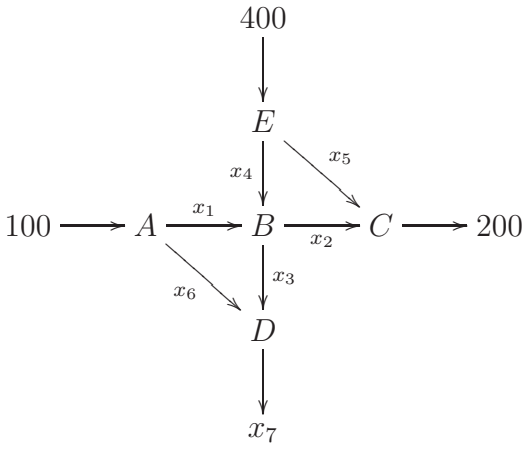
Is the vector $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? You should justify your answer.

5. **(2 points)** For what value(s) of h are the vectors

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ -7 \\ -6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ h \end{bmatrix}$$

linearly independent? You should justify your answer.

6. **(5 points)** Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system at this stage.)

(b) The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 100 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 400 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

(c) Suppose that due to road work, the flow along AD is limited to a maximum of 50 cars per minute. What is the minimum possible flow along BD?

This page is only for scrap work and is intentionally left blank.